

EMERGENT VEGETATION



FISH EAGLES use their long talons to snatch fish from the surface of the water.



THE JACANA, or lilytrotter, has long toes that spread the bird's weight so that it can stand on floating leaves.



THE CAPYBARA, the world's largest rodent, feeds on land, but uses water as a refuge from attack by predators.

SURFACE



CAIMANS, and many other crocodilians, can survive for many weeks when wetlands dry out by wallowing in moist mud.

MIDWATER



THE COMMON SNAKE-NECKED TURTLE can stay underwater for long periods due to its slow metabolism.



LUNGFISH survive times of drought by lying dormant in mud, protected by their waterproof cocoons.

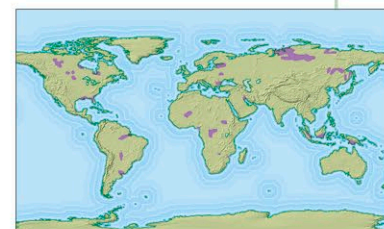
BOTTOM

Wetland

A wetland is any waterlogged or flooded area with a covering of water plants. In some wetlands—reedbeds and bogs, for example—the plants hide the water completely. However, in most wetland, areas of open water and dense vegetation are mixed, creating a rich and complex habitat that can be exploited by animals of almost every kind.

Biologically, wetland is among the most productive inland habitats, sometimes surpassing even rain forest in the amount of food that it generates for animals. In temperate parts of the world, this productivity reaches a peak during spring and summer, but in the tropics and subtropics, it is more affected by the water supply. Some tropical wetlands—South America's Pantanal, for example—largely dry out during the dry season, but then look like vast lakes once it has rained.

In many wetlands, the water is no more than a yard (3¼ ft) or so deep, which means that bottom-living animals and surface dwellers are rarely far apart. This kind of environment is ideal for air-breathing swimmers, such as snakes and turtles, and also for land-based animals that use water as a temporary refuge from danger. Unlike large lakes, wetlands have an extra dimension in the form of emergent plants, which grow up through the water's surface and into the air. These plants range in size from small grasses and rushes just tall enough to keep insects clear of the water, to water-loving trees that grow to over 115 ft (35 m) high. Trees act as important roosting and breeding sites for waterbirds, providing them with shelter and relative safety from predators as well as keeping them close to the source of their food.



FRESHWATER WETLANDS include the Pantanal, South America, and the Okavango, an inland delta in southern Africa.

POLLUTION AND ABSTRACTION

Freshwater habitats are affected by human activities in many ways. Water pollution is a general problem that has profound consequences for a wide range of animal life. However, in addition, large areas of wetland have been drained for agricultural use, often because wetland conditions produce highly fertile soil. Water abstraction, both for agriculture and for domestic use, is also an increasing threat to freshwater habitats. Efforts to protect

wetlands have produced several international agreements. One of these, the Ramsar Convention, adopted in 1971, focuses on wetlands used by migrating waterfowl.



HIGH AND DRY In central Asia, the Aral Sea—a giant inland lake—has shrunk dramatically because incoming river water has been diverted to irrigate fields.